Public GEO Lung Cancer Expression Analysis - GSE10072

Ahmed Tharwat Hassan | Junior bioinformatics portfolio project | Source: NCBI Gene Expression Omnibus

Objective

Compare processed microarray expression values between lung tumor and non-tumor tissue samples from public GEO series GSE10072, then summarize the strongest probe-level differences using a reproducible Python workflow.

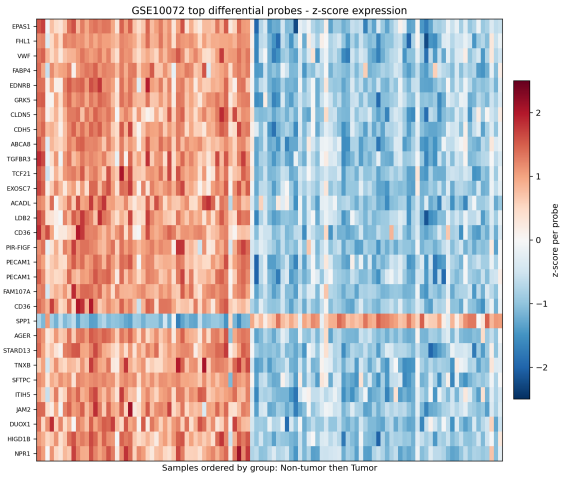
Dataset and Methods

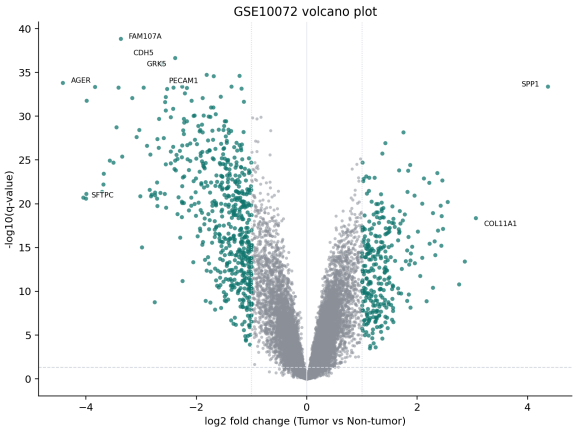
Dataset: GSE10072 (58 tumor samples, 49 non-tumor samples). Analysis used pandas, SciPy, and matplotlib. For each probe, I computed group means, Tumor vs Non-tumor log2 fold change, Welch two-sample t-test p-values, and Benjamini-Hochberg FDR q-values. Significant probes were flagged at $q < 0.05$ and absolute log2FC >= 1.0.

Results

The workflow identified 859 significant probes by the stated threshold. The table below lists the top probes by q-value; symbols are shown when available from the GPL96 platform annotation.

Probe	Gene	Non-tumor	Tumor	log2FC	p-value	q-value
209074_s_at	FAM107A	10.82	7.45	-3.37	6.64e-44	1.48e-39
204396_s_at	GRK5	9.97	7.58	-2.39	2.01e-41	2.24e-37
204677_at	CDH5	9.42	6.80	-2.62	1.32e-40	9.78e-37
208982_at	PECAM1	11.65	9.84	-1.81	3.36e-39	1.87e-35
213103_at	STARD13	8.85	7.63	-1.22	5.22e-39	2.33e-35
208981_at	PECAM1	11.07	9.38	-1.69	7.22e-39	2.68e-35
210081_at	AGER	11.88	7.46	-4.42	4.77e-38	1.52e-34
202112_at	VWF	11.83	9.57	-2.26	1.46e-37	3.97e-34





Interpretation and Limitations

This is a compact public-data demonstration of expression analysis, not a clinical diagnostic claim. It shows data access, metadata parsing, differential-expression statistics, FDR correction, visualization, and written interpretation. Because the source is microarray data, results are reported at probe level and should be interpreted with platform annotation context.